Math 141, F11 Name: Answer

Quiz 6 Section: SSN: xxx-xx-

<u>Instructions:</u> There is totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(6 points) Find the derivatives of the following functions.

(a) $f(t) = t^2 \sin t$

$$f'(t) = t^2 \frac{d}{dt}[\sin t] + \sin t \frac{d}{dt}[t^2]$$
$$= t^2 \cos t + 2t \sin t$$

(b) $h(x) = e^{\tan x}$

$$h'(x) = e^{\tan x} \frac{d}{dx}[\tan x]$$
$$= e^{\tan x} \sec^2 x$$

(c) $g(x) = \left(\frac{x+1}{x-1}\right)^8$

$$g'(x) = 8\left(\frac{x+1}{x-1}\right)^7 \frac{d}{dx}\left[\frac{x+1}{x-1}\right]$$
$$= 8\left(\frac{x+1}{x-1}\right)^7 \frac{(x-1)\frac{d}{dx}[x+1] - (x+1)\frac{d}{dx}[x-1]}{(x-1)^2}$$
$$= 8\left(\frac{x+1}{x-1}\right)^7 \frac{-2}{(x-1)^2}$$
$$= \frac{-16(x+1)^7}{(x-1)^9}$$

Problem 2.(4 points) Use **implicit differentiation** to find dy/dx of

$$x^3 + 2y^3 = 1.$$

$$\frac{d}{dx}[x^3 + 2y^3] = \frac{d}{dx}[1]$$
$$3x^2 + 6y^2 \frac{dy}{dx} = 0$$
$$6y^2 \frac{dy}{dx} = -3x^2$$
$$\frac{dy}{dx} = -\frac{x^2}{2y^2}$$